UTPAL SONOWAL

+91 9864341272 | sonowalu73@gmail.com | Linkedin | Github | Portfolio

SUMMARY

Full-stack developer building responsive interfaces, secure APIs, and AI-powered applications using React, Next.js, TypeScript, Node.js, and PostgreSQL, with Docker and AWS deployment experience.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C/C++
Frontend: React, Next.js, React Router, Redux Toolkit, Context API, Axios
Backend: Node.js, Express.js, Fast API, JWT Authentication, Authentication & Authorization, Zod
Databases: SQL**,** PostgreSQL, MongoDB, Prisma ORM, Redis
AI/ML: LLMs, RAG, Prompt Engineering, Gemini API, Embeddings, Vector Databases
DevOps: AWS (EC2, S3, Lambda), Docker, CI/CD, Github Actions, Vercel, Render
Tools: Git, GitHub, Postman, Figma, VS Code, K6

EXPERIENCE

Full Stack Developer Intern *Jan 2025 – Dec 2025*
National Informatics Centre (NIC) — India (Laravel, PHP, PostgreSQL, JavaScript, Bootstrap)

- Developed a two-tier transportation management system to automate transportation billing across warehouses, wholesalers, retailers, and commodity distribution.
- Automated transportation cost and billing calculations based on commodity, distance, and geographical area, eliminating complex manual calculations**.**
- Reduced billing preparation time from approximately 3–4 hours to around 10 minutes by automating data processing and bill generation.
- Generated downloadable Excel billing sheets with automated formatting based on predefined templates and designs, eliminating repetitive manual spreadsheet preparation.

PROJECTS

LnkShrt — Next Gen URL Shortener & Analytics Live Demo | GitHub
Node.js, Express, PostgreSQL, Redis, TypeScript, Next.js, Docker, AWS(EC2), Nginx, GitHub Actions

- Built a full-stack URL management platform using Next.js, TypeScript, and Express, supporting custom aliases, password protection, expiration, UTM parameters, and QR codes to centralize link management.
- Implemented custom-domain ownership verification using DNS TXT and CNAME checks, enabling users to create branded short links while restricting domain assignment to verified owners.
- Deployed the backend and Redis on AWS EC2 using Docker, Nginx, and Let's Encrypt HTTPS; automated TypeScript, API, database, and container checks through GitHub Actions to catch errors before deployment.
- Load-tested the URL redirect API using k6 (100 VUs, 18,000 requests): 20.68 ms p95 latency at 200 RPS, 0% HTTP failures.

Study Buddy — Full-Stack RAG Chatbot with Production-Grade Authentication Live Demo | Github
Fast API, PostgreSQL, Pgvector , Redis, Gemini API, TypeScript, React.js, Docker,

- Designed and built a production-grade auth system with email/OTP + Google/GitHub OAuth, including account linking, email verification.
- Implemented secure session management using access/refresh token rotation with reuse detection, HttpOnly/Secure cookies, and full session revocation.
- Developed a RAG pipeline (PostgreSQL + Pgvector) enabling users to query their personal notes with cited, source-linked LLM-generated answers.
- Enforced multi-tenant data isolation so retrieval and queries never cross user boundaries.
- Added Redis-backed rate limiting and **19 automated backend tests** covering authentication, retrieval, streaming, and failure recovery.
- Built background job infrastructure (Redis) for async embedding generation and retry-safe transactional email delivery.

EDUCATION

Bachelor of Computer Applications (BCA) 2023 – 2026
North Lakhimpur University
CGPA: 7.4 / 10